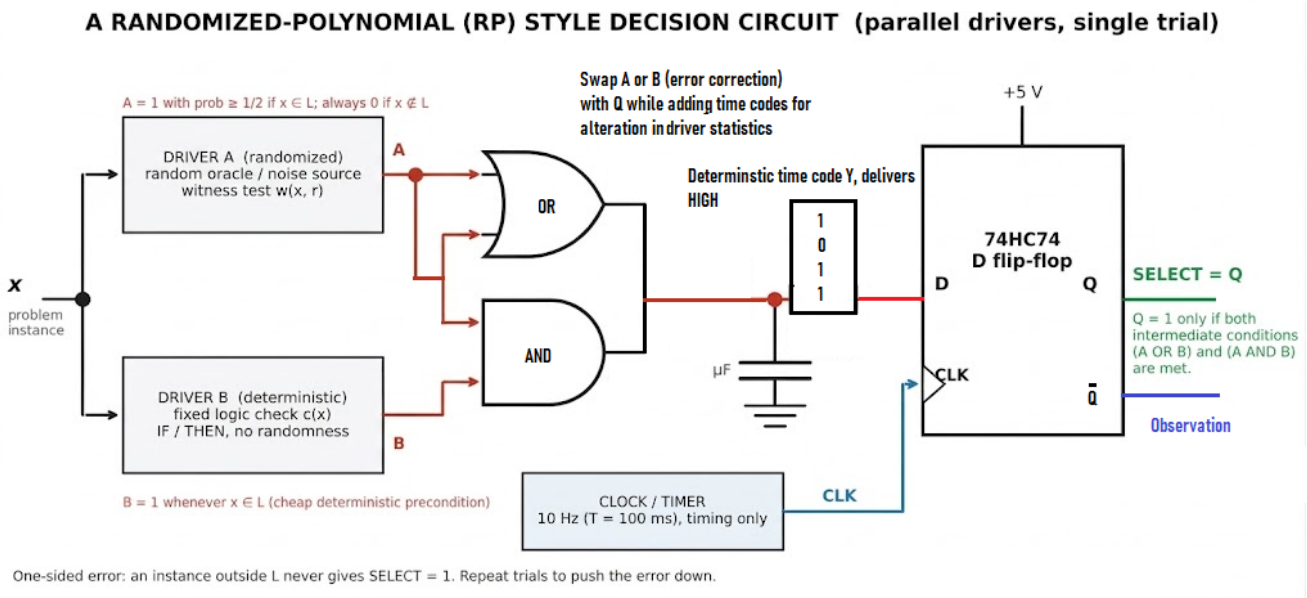
How the RP-Style Decision Circuit Works

A walkthrough of the parallel-driver, single-trial circuit (A, B, OR/AND, time code Y, 74HC74 flip-flop)



The circuit as supplied

1. The big idea

The circuit is a hardware picture of a **Randomized Polynomial-time (RP)** decision procedure. RP is the class of yes/no problems for which a probabilistic algorithm has **one-sided error**: if the answer is **NO** (x is not in L) the algorithm *never* says yes, and if the answer is **YES** (x is in L) it says yes with probability at least 1/2. The circuit runs one such trial. It takes the problem instance **x**, runs it through two drivers in parallel, combines their outputs with logic gates, and stores the verdict in a flip-flop on a clock edge. That stored bit, **SELECT = Q**, is the answer.

2. The blocks and what each one does

Block	Role	Behaviour
Driver A (randomized)	Probabilistic witness test	Uses a random oracle / noise source to try a random witness r and test w(x, r). If x is in L, A = 1 with probability ≥ 1/2. If x is not in L, A = 0 always (no witness can exist).
Driver B (deterministic)	Cheap precondition	A fixed IF/THEN check c(x) with no randomness. B = 1 whenever x is in L. It acts as a sanity filter that costs almost nothing.
OR gate	“Swap / error-correction” branch	Outputs 1 when A or B (or both) is 1, so either driver can contribute. In the figure's notes it is the stage that swaps A or B and adds time codes to alter the driver statistics.
AND gate	Agreement check	Outputs 1 only when A = 1 and B = 1, i.e. the random witness test succeeded *and* the deterministic check agrees.
Capacitor (μF) node	Smoothing / hold	The gate outputs meet at a node with a capacitor to ground. It low-pass filters glitches so the signal is stable when the clock samples it.
Time code Y = 1011	Deterministic time code	A fixed bit pattern that delivers a HIGH to the D input, giving the stage a known, repeatable timing signature.
Clock / timer	Sampling instant	10 Hz (T = 100 ms), “timing only”. It carries no data, it just tells the flip-flop when to look at D.

Block	Role	Behaviour
74HC74 D flip-flop	Memory of the verdict	On the rising CLK edge it copies D to Q and holds it. Q is SELECT (the decision). Q̄ (Q-bar) is the inverted output, labelled Observation.

3. Step-by-step signal flow

1. **Input.** The instance x is fanned out to both drivers at once (the black dot at the left).
2. **Drivers fire in parallel.** A flips its random coin(s) and tests a witness. B runs its fixed check. Both produce one bit, labelled A and B in red.
3. **Gates combine them.** A and B go to the OR and AND gates. The caption summarises the requirement as two intermediate conditions: (A OR B) and (A AND B).
4. **Node + capacitor.** The gate outputs join at one node. The capacitor smooths it and the time code Y presents a defined HIGH to the flip-flop's D pin.
5. **Clock edge.** Every 100 ms the rising edge on CLK makes the 74HC74 latch whatever is on D.
6. **Result.** Q = SELECT = 1 means “accept: x is in L”. Q = 0 means “no witness found”. Q-bar shows the opposite polarity for observation.

4. Why the error is one-sided

The caption says Q = 1 only if both conditions hold, which collapses to **A AND B** (since A AND B already implies A OR B). So the circuit accepts only when the randomized driver finds a witness:

Case	A	B	SELECT (Q)	Meaning
x not in L	always 0	any	0	Never a false accept. No error possible.
x in L, witness found	1	1	1	Correct accept (probability ≥ 1/2).
x in L, witness missed	0	1	0	False reject (probability ≤ 1/2). The only error type.

This matches the footer of the diagram: *an instance outside L never gives SELECT = 1*. A wrong answer can only be a missed YES, never a fake YES.

5. Repeating trials to push the error down

One trial can wrongly reject a YES instance with probability up to 1/2. Because trials are independent and a single success is conclusive, you run the circuit k times and accept if *any* trial gives SELECT = 1. The chance of every trial failing is at most $(1/2)^k$. At the 10 Hz clock, k trials take k × 100 ms.

Trials k	Worst-case error	Time at 10 Hz
1	0.5	0.1 s
2	0.25	0.2 s
5	0.03125	0.5 s
10	0.000976562	1.0 s
20	9.54e-07	2.0 s

6. Things to be aware of in the drawing

The figure is a conceptual illustration, not a build-ready schematic. If you ever wire it up, check these points:

- **How the two gates combine.** AND implies OR, so requiring both conditions, (A OR B) and (A AND B), reduces to A AND B. Tied together as drawn, the two outputs would instead behave like a plain OR (A OR B). The caption's “both conditions” reading is the one that keeps the logic

meaningful.

- **Tied gate outputs.** Directly joining two CMOS outputs causes bus contention when they disagree. A real design would use an AND gate (or a proper combining stage) to merge them.
- **A OR B alone would break the guarantee.** If B can be 1 for an x outside L, accepting on A OR B could give a false accept. Only the AND path preserves one-sided error.
- **74HC74 housekeeping.** Real use needs the PRE and CLR pins tied high, VCC and GND connected, and decoupling near the chip. These are not shown.
- **Q-bar is not extra information.** It is simply NOT Q, useful for an indicator LED or active-low logic.

Summary: two drivers test x in parallel, the gates demand agreement, a clocked D flip-flop latches the verdict, and repetition drives the false-reject rate toward zero while false accepts stay impossible.